METRIC

A-A-52407
September 15, 1992

COMMERCIAL ITEM DESCRIPTION

GLASS, FILTER, COLORED (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-G-47034A, which is canceled.

1. Abstract. This CID covers the material requirements and physical properties of four related grades of colored filter glass for use in optical systems. The colored filter glass is referred to herein as "glass".

1.1 Classification. Glass covered by this CID shall be classified as follows:

Grade A - Sharp cut red filter.
Grade B - Sharp cut red filter with lower cut frequency than grade A glass.
Grade C - Blue filter.
Grade D - Blue filter which passes smaller frequency range between 15% and 60% transmittance in blue portion of spectrum than grade C glass.

2. Salient characteristics.

2.1 Materials. The materials shall be of high quality silica, boric oxide, soda, aluminum oxide, colorants, and other ingredients necessary to produce fine annealed high-quality glass which conforms to the requirements for grade as detailed in this CID. The use of recovered material made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.5).

2.2 Design and construction. The glass shall be designed and constructed in accordance with the applicable engineering drawing (see 5.2).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 9340

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2.2.1 Control thickness. The glass shall meet the control thickness requirements specified in table I. The molder/blank thickness shall be as required to achieve the control thickness and salient characteristics of 2.

TABLE I. Control thickness.

Grade	Control thickness, mm
A	1.4 - 4.6
B	1.4 - 4.6
C	3.9 - 5.1
D	4.4 - 5.6

2.2.2 Inclusions. The allowable maximum and minimum size inclusions shall be 0.50 millimeter (mm) and 0.10 mm, respectively. The permissible number of maximum size inclusions shall be one per cubic centimeter (cc) of glass. The sum of the diameter of all inclusions larger than the minimum size per cc of glass shall not exceed the diameter of the allowable maximum size. Inclusions smaller than the minimum size shall be disregarded.

2.3 Physical properties.

2.3.1 Coefficient of expansion. The glass shall meet the coefficient of expansion requirements specified in the applicable engineering drawing when tested in accordance to ASTM E228.

2.3.2 Refractive index. The glass shall meet the refractive index requirements specified in the applicable engineering drawing when tested in accordance to ASTM D542.

2.3.3 Temperature. The glass shall meet the maximum annealing point temperature and minimum softening point temperature requirements specified in the applicable engineering drawing when tested in accordance with ASTM C336 and ASTM C338, respectively.

2.3.4 Specific gravity. The glass shall meet the specific gravity requirements specified in the applicable engineering drawing when tested in accordance with ASTM D792.

2.3.5 Transmittance wave length. The glass grades A and B (sharp cut red filters) and grades C and D (blue filters) shall meet the transmittance wave length requirements specified in tables II and III respectively.

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TABLE II. Transmittance wave length (grades A and B).

Grade	Nanometers			
	1/	2/	3/	4/
A	637 - 648	678	617	15
B	628 - 637	667	608	15

1/ The wave length limits within which shall lie the wave length (the cut frequency) where the transmittance is 37 percent (%).
2/ The transmittance shall be 80% or greater from this wave length to 750 nanometers.
3/ The transmittance shall be less than 0.5% at all shorter wave lengths.
4/ The difference between the wave lengths where transmittance is 15% and 60% is not greater than this value.

TABLE III. Transmittance wave length (grades C and D).

Grade	Nanometers		
	1/	2/	3/
C	555	49	T 740 = T 530 ± 15
D	514	40	T 740 = T 510 ± 5

1/ Wave length at which transmission is 15.0 ± 0.7%.
2/ The difference between the wave lengths at which the transmission is 15% and 60% shall be equal to the specified value within a tolerance of ± 5 millimicrons. This requirement is applicable in the blue portion of the spectrum.
3/ Transmittance (T) in % not corrected for surface reflection.

2.4 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform and in conformance with this CID. The glass shall be free of dirt, foreign materials, cracks, or contaminants.

3. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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ated, double synthetic jacket, with

4. Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.2).

5. Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Addresses for obtaining copies of referenced Non-Government publications. American Society for Testing and Materials (ASTM) C336 "Method for Annealing Point and Strain Point by Fiber Elongation"; ASTM C338 "Softening Point of Glass"; ASTM D542 "Index of Refraction of Transparent Organic Plastics"; ASTM D792 "Specific Gravity and Density of Plastic by Displacement"; and ASTM E228 "Linear Thermal Expansion of Rigid Solids with Vitreous Silica Dilatometer" are available from American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of the Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of the documents referenced (see 5.1).
c. Grade of glass (see 1.1).
d. Drawing number, revision letter and date of the applicable engineering drawing, and the end item part number.
e. Level of preservation, packaging and packing (see 4).

5.3 Cross-reference data. Glass conforming to this CID is substitutable with colored filter glass conforming to MIL-G-47034A.

5.4 Definitions.

5.4.1 Control thickness. The control thickness is the allowable thickness within the specified range necessary to meet the requirements of a given glass grade.

5.4.2 Cut frequency. The frequency within a given range that corresponds to a desired transmittance change around a specified transmittance value.

5.5 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Custodians:
Army - AT
Navy - AS
Air Force - 99

Review activities:
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Preparing activity:
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